

01:XXX:XXX - Homework n

Pranav Tikkawar

March 24, 2026

1. Notion, process are not equally smooth in time and space. Simple case 1d space and time. If they have different smoothness. $(\alpha^2 + \Delta)^\beta Z(x, t) = w(x, t)$

Come to an analog that works numerically, Very small number of models that have a closed form for covariance.

$\frac{1}{(\alpha^2 + |\omega|^2)^2 + \nu^2}$ JASA 2005. Search for this model and ppl have done it and it is similar to the heat equation.

Propose an algorithm, and then ask how well it does, since I have a closed form

Find the Software. Rue, Lindgren.

inla software